Data Source Evaluation Report

How ready your sources were, and what was left unmapped

Date: 2026-10-06

Project: test-claude-26 (Customer 360, bootcamp-generated synthetic data)

Summary

- Total data sources: 3
- Entity Specification-compliant: 0
- Needs mapping: 3
- Needs enrichment: 0

Data Source Details

Data Source 1: CRM_CONTACTS

Status: Needs mapping

Location: data/raw/crm_contacts.csv

Records: 4,000

Fields: 6 columns

Evaluation:

- contact_id -> RECORD_ID; full_name -> NAME_FULL (one name field, so a direct mapping); email -> EMAIL_ADDRESS; phone -> PHONE_NUMBER; address -> ADDR_FULL (one free-text address field).
- created_date has no Entity Specification counterpart (a candidate for payload or ignore).
- No RECORD_TYPE field; DATA_SOURCE is derivable (CRM_CONTACTS).
- Quality 93.4 (completeness 93.3, format 92.2, duplicates 0.0).

Reason: Raw export with non-specification field names.

Record type check: 0 of 0 PERSON-typed records are type/name candidates. Decision: none needed

Next step: Phase 2 (mapping)

Data Source 2: WEB_ORDERS

Status: Needs mapping

Location: data/raw/web_orders.csv

Records: 4,500

Fields: 11 columns

Evaluation:

- order_id -> RECORD_ID; first_name/last_name -> NAME_FIRST/NAME_LAST; email -> EMAIL_ADDRESS; phone -> PHONE_NUMBER; street/city/state/zip -> ADDR_LINE1/ADDR_CITY/ADDR_STATE/ADDR_POSTAL_CODE.
- order_date and order_total have no counterpart.
- No RECORD_TYPE field; DATA_SOURCE is derivable (WEB_ORDERS).
- Quality 89.5 (completeness 86.0, format 97.1, duplicates 0.0).

Reason: Raw export with non-specification field names.

Record type check: 0 of 0 PERSON-typed records are type/name candidates. Decision: none needed

Next step: Phase 2 (mapping)

Data Source 3: SUPPORT_TICKETS

Status: Needs mapping

Location: data/raw/support_tickets.jsonl

Records: 1,500

Fields: 8 keys

Evaluation:

- ticket_id -> RECORD_ID; requester_name -> NAME_FULL; requester_email -> EMAIL_ADDRESS; phone -> PHONE_NUMBER; city -> ADDR_CITY; postal_code -> ADDR_POSTAL_CODE.
- topic and opened_on have no counterpart.
- No RECORD_TYPE field; DATA_SOURCE is derivable (SUPPORT_TICKETS).
- Quality 75.5 (completeness 66.5, format 95.8, duplicates 0.0); the gaps are deliberate in this generated source.

Reason: Raw export with non-specification field names; sparse.

Record type check: 0 of 0 PERSON-typed records are type/name candidates. Decision: none needed

Quality gate (70-79): the Bootcamper chose to continue to mapping, accepting the documented gaps: postal_code missing in 46% of records, city 50%, phone 49% (deliberate in this generated source); off-pattern phone (71) and opened_on (109) values left as received.

Next step: Phase 2 (mapping)

Mapping Priority

1. WEB_ORDERS - largest source, with parsed names and addresses. 2. CRM_CONTACTS - strong quality; exercises single-field name and address mapping. 3. SUPPORT_TICKETS - sparsest; benefits most from the other sources being loaded first.

Cross-Source Outlook

- CRM_CONTACTS x WEB_ORDERS - measured

    - Shared attribute: EMAIL_ADDRESS
    - Distinct values shared: 1,404 (of 3,531 in A, 4,122 in B); phone digits shared: 929
- WEB_ORDERS x SUPPORT_TICKETS - measured
    - Shared attribute: EMAIL_ADDRESS
    - Distinct values shared: 481 (of 4,122 in A, 1,207 in B); phone digits shared: 222
- CRM_CONTACTS x SUPPORT_TICKETS - measured
    - Shared attribute: EMAIL_ADDRESS
    - Distinct values shared: 390 (of 3,531 in A, 1,207 in B); phone digits shared: 224

Quality visual: docs/visualizations/data_quality_assessment.html